University of Aveiro, PRI, 2025

Assignment 2

In Assignment 1, you implemented a classic Information Retrieval system based on term-matching (BM25). **Assignment 2 focuses on upgrading your system with "AI" capabilities.** You will transition from a purely lexical search engine to a **Semantic Search System** by integrating Neural Networks.

This work is of an exploratory nature. You are not expected to train models from scratch. Instead, you are encouraged to research, select, and integrate existing pre-trained models (Zero-Shot learning) and APIs to enhance your system.

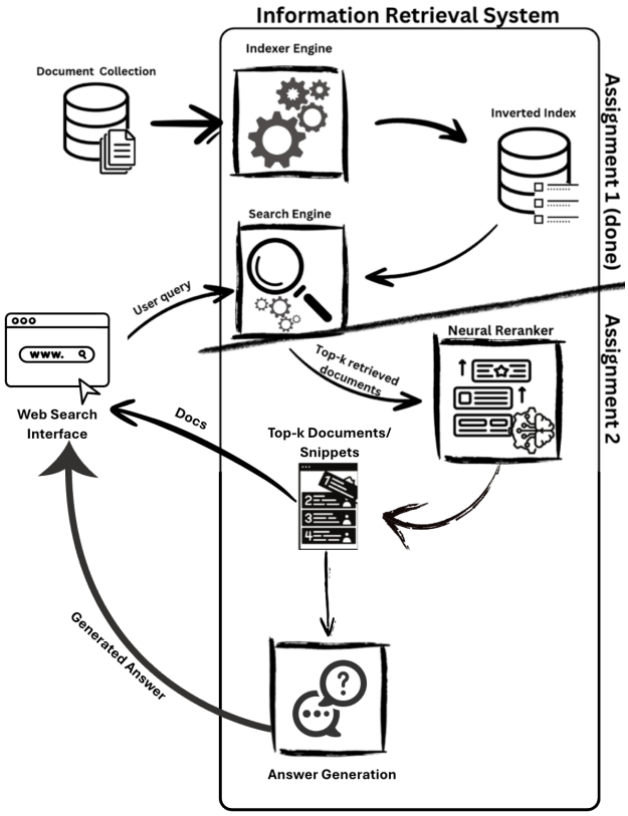

Figure 1: Overview of the components in an Information Retrieval System.

Core Tasks:

1. **Neural Reranking:** Improve the ranking quality of the results retrieved by BM25.
2. **Answer Generation (RAG):** Provide natural language answers conditioned on the retrieved documents.
3. **Further AI Enhancements (Optional):** Explore additional capabilities like semantic snippets or embedding-based feedback.

1. Neural Reranker (10)

The goal of the Neural Reranker is to re-order the candidate documents retrieved by your Search Engine (Assignment 1) to improve precision at the top of the list (e.g., Top-10).

How it works: Instead of scoring every document in the collection (which is too slow), you will implement a two-stage retrieval pipeline:

1. **Retrieve:** Use your BM25 indexer to fetch the top-K candidates (e.g., K<100).
2. **Rerank:** Pass the query and these K documents into a **Neural Relevance Model** to calculate a semantic similarity score.
3. **Sort:** Re-order the documents based on the neural score.

Where to start:

- https://huggingface.co/cross-encoder/mmarco-mMiniLMv2-L12-H384-v1
- https://huggingface.co/unicamp-dl/mMiniLM-L6-v2-pt-v2

2. Answer Generation (7)

Provide a conditioned generated answer on top of the retrieved documents.

How it works:

1. Take the Top-1 document or snippet (after neural reranking).
2. Construct a prompt that includes the user's question and the document content.
3. Send this prompt to a Large Language Model (LLM) to generate a final answer.

Where to start:
- https://ai.google.dev/gemini-api/docs/rate-limits
- https://console.groq.com/docs/rate-limits

3. Further AI Enhancements (3)

To achieve the highest grade, you are encouraged to explore additional AI-driven features. This is your chance to be creative.

Ideas for exploration:

- **A. Semantic Snippet Extraction:** Use a relevance model to find the most relevant paragraphs and show those in the results page.
    - *Tip:* You can use the same Cross-Encoder from step 1 to score individual paragraphs within the document.
- **B. Embedding-Based Relevance Feedback:** Improve the "More Like This" feature from Assignment 1.

    - *Tip:* Compute embeddings (vectors) for the documents. If a user likes Document A, perform a Vector Search (Cosine Similarity) to find documents with similar embeddings, rather than just similar words. NOTE: THIS REQUIRES A HUGE PRECOMPUTATION ON GPU. Use Colab or Kaggle.
- **C. Query Expansion:** Use an LLM to rewrite the user's query before searching.

4. Submission Requirements

- A README file that details what was implemented and **all of the design choices**.
    1. What/How much information are you feeding to the LLM
    2. What type of model are you using as a reranker and why? Did you explore alternatives?
    3. Any additional AI enhancement?
- All developed code must be in a runnable state (We must be able to run your code without issues). When using LLM APIs, make your code in a way that the API KEY is always obtained from env variable and use standard ones.

5. Evaluation Criteria

Your assignment will be evaluated based on the following criteria:

- Work progression along time
- Presentations
- AI components
    1. The model choices are adequate and make sense.
    2. The enhancements clearly improve user experience.